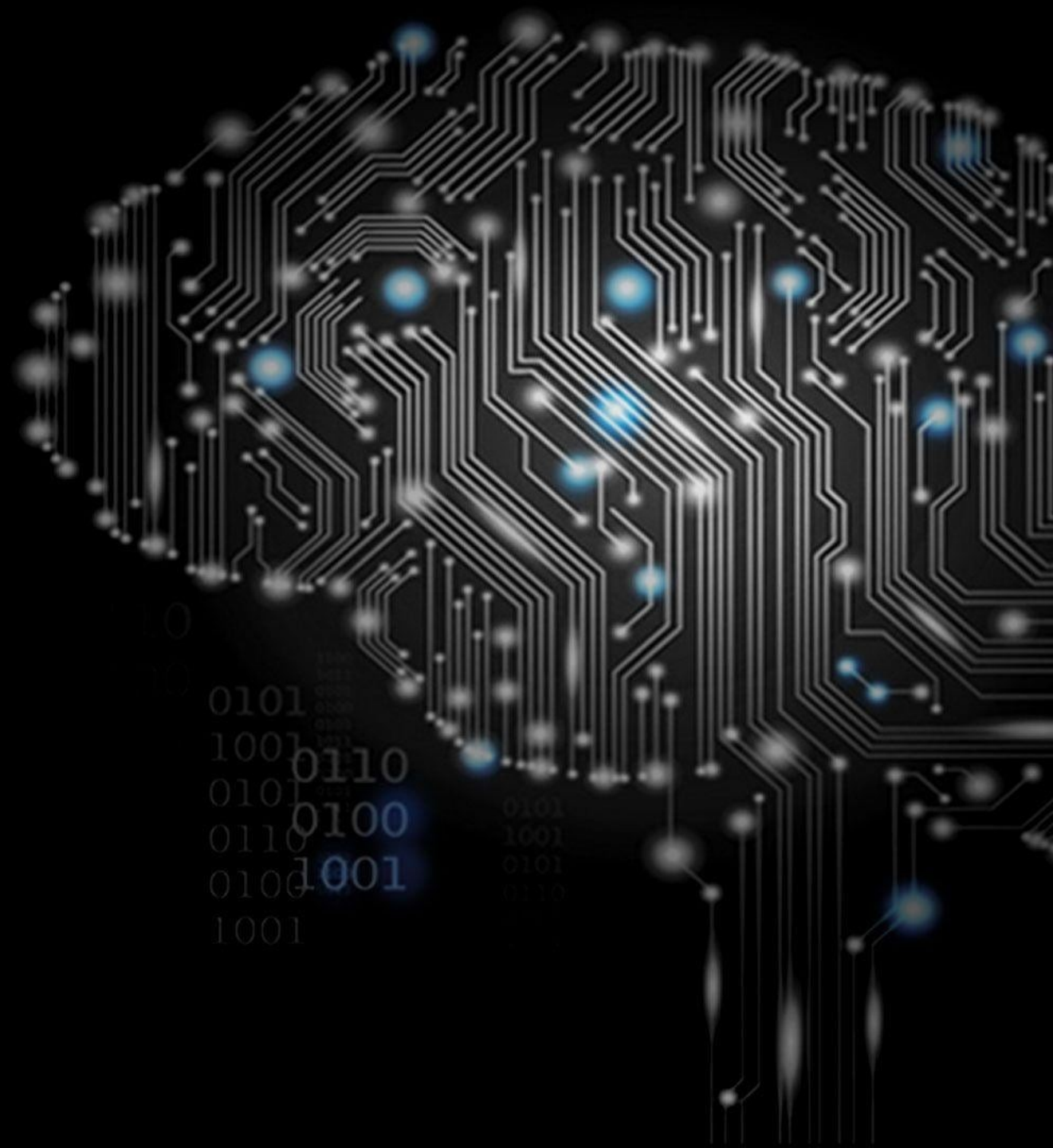

Introduction to Generative AI

3. Diffusion Models

Module Topics

- Generative adversarial networks (GANs)
- Autoencoders and Variational Autoencoders
- Diffusion Models
- Multi-modal Models
- CLIP

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Generative Adversarial Networks

- GANs were introduced in 2014 and revolutionized image generation
- Two neural networks compete
  - Generator which tries to create fake images
  - Discriminator which tries to detect fakes
  - They train in an adversarial game
    - Generator improves by fooling the discriminator
    - Discriminator improves by detecting fakes
- Over time, the generator becomes very good at creating realistic images

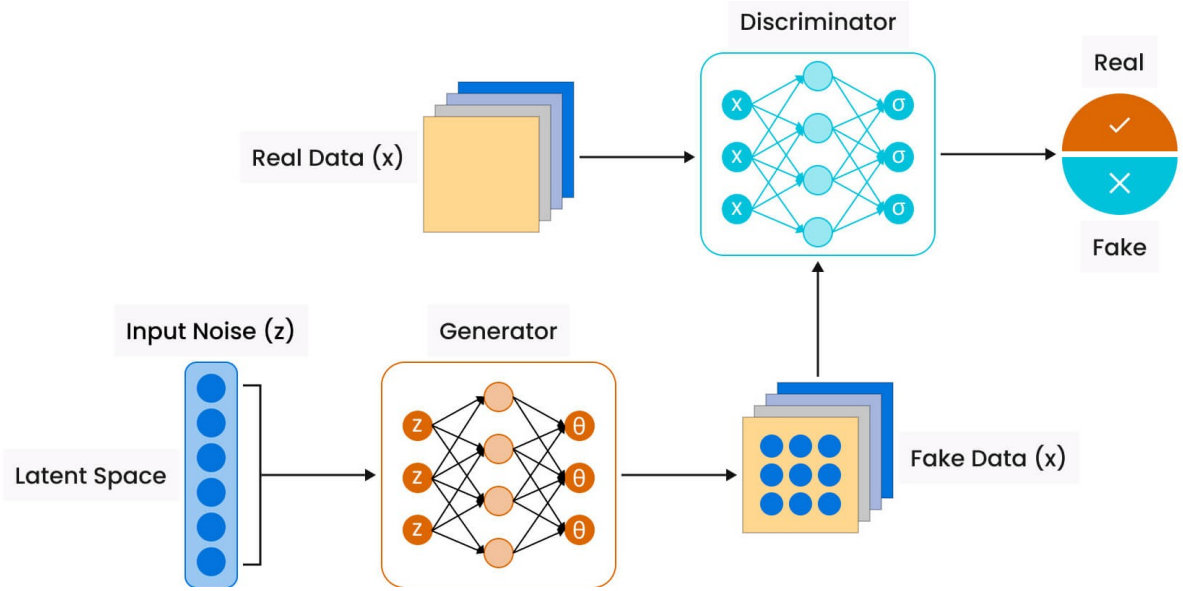

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Image Credit:https://www.solulab.com/generative-adversarial-network/

Generative Adversarial Networks

- Why GANs were significant
  - Produced the first truly photo-realistic synthetic faces
  - StyleGAN produced high-resolution human faces
  - Enabled deepfakes
  - Opened the door to synthetic media
- The latent space
  - A structured space of abstract features from which the generator constructs images
  - The model created from training on images
  - More on this later

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Model Collapse

- Refers to a failure mode where
  - The generator produces very limited varieties of outputs, even though the real data distribution is diverse
- Reasons for model collapse
  - Adversarial instability
    - Two networks competing can create unstable dynamics
  - Feedback imbalance
    - If discriminator gets too strong or too weak, training destabilizes
  - Local optima
    - Generator finds a narrow region that works and stops exploring
  - No explicit diversity objective
    - The GAN loss does not directly enforce output diversity

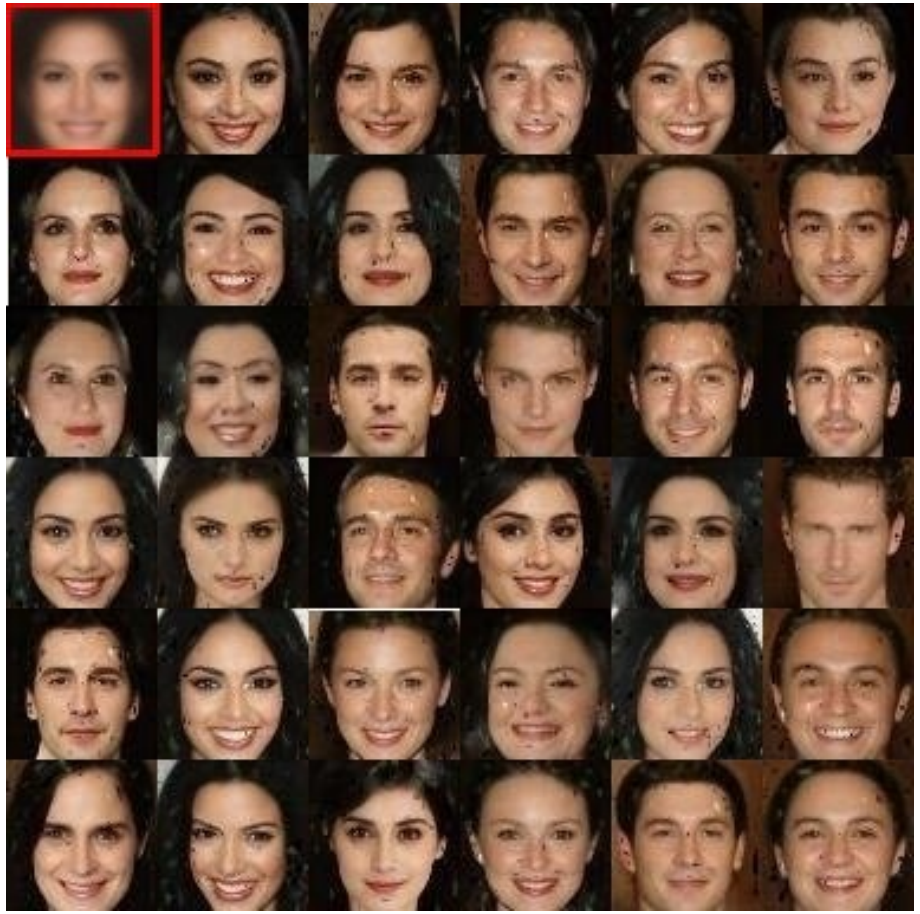

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Image Credit https://www.researchgate.net/figure/Examples-for-mode-collapse-in-BEGAN-trained-on-CelebA-showing-three-over-allocated-bins_fig2_325484013

Autoencoders

- The encoder learns two things
  - A encoding function that maps a data point (image) to an encoded representation
  - A decoder that reconstructs a data point from an encoded representation
- The objective is to learn an efficient feature representation
  - Learning takes place by minimizing the reconstruction loss or how different the reconstructed data is from the original data
- The autoencoder's task is to understand only the deep correlations and relationships among the data
  - Done by forcing dimensionality reduction to force the decoder to actually have to reconstruct the data
  - Otherwise, the trivial autoencoder would just return the original data point unmodified

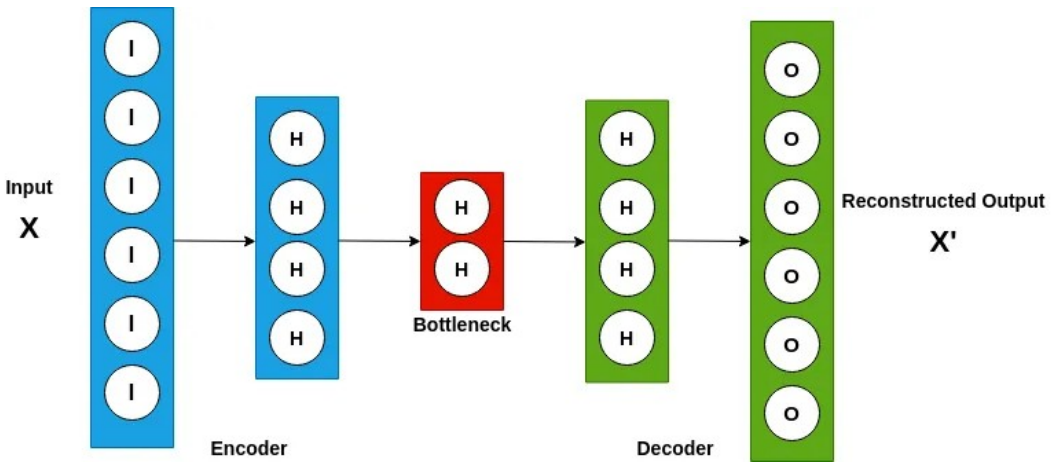

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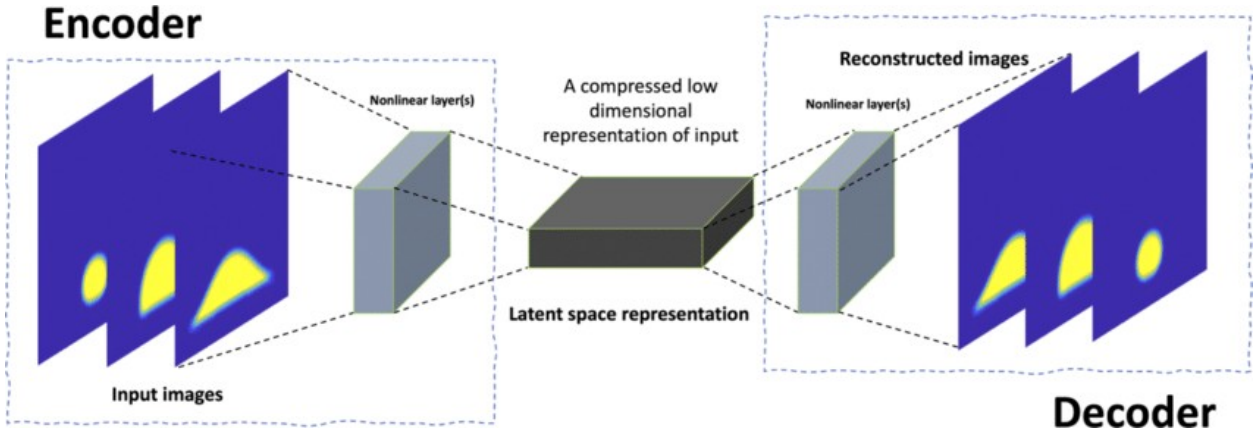
Image Credit https://medium.com/data-science/introduction-to-autoencoders-7a47cf4ef14b

Latent Space

- Uses a bottleneck layer
  - The number of nodes in the bottleneck layer gives the dimension of the encoding of the input
- The representation from the bottleneck layer is called the latent space
  - This forces the model to explore relationships among the features
  - Exploits the natural structure of the data
  - Requires that relationships do exist
    - No encoding can be done if all the features are independent
  - Can learn non-linear relationships
  - Should be sensitive enough to minimize reconstruction loss but not to overfit the data
- Back propagation is used to train the encoder



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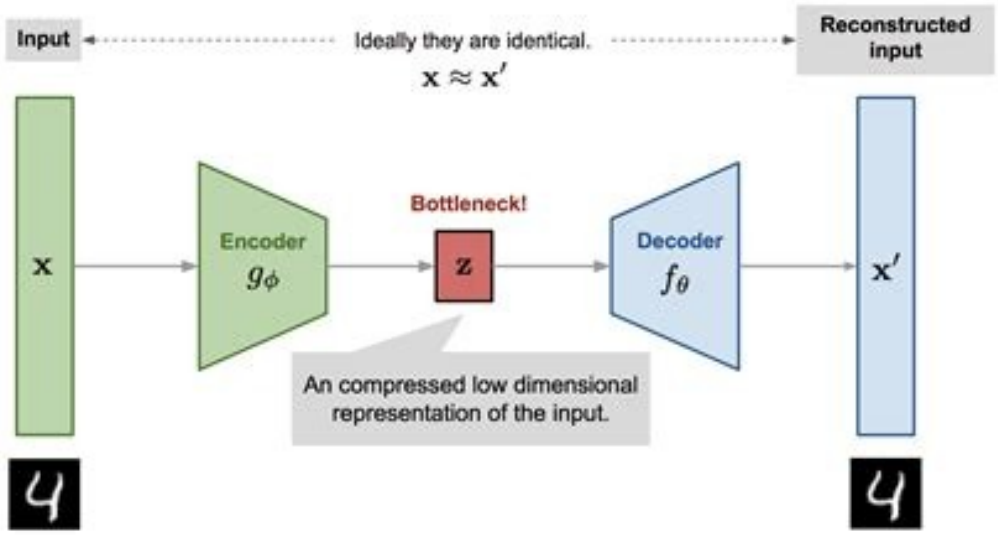
The Encoder

- Takes high-dimensional input and compresses it into a lower-dimensional vector
  - Extracts essential features
  - Mathematically where “g” is the encoder function
    - z=b(x)
    - x = input (image). z = latent representation
- It learns to discard
  - Pixel noise
  - Redundant details
- And preserve
  - Structure
  - Shape
  - Important semantic features



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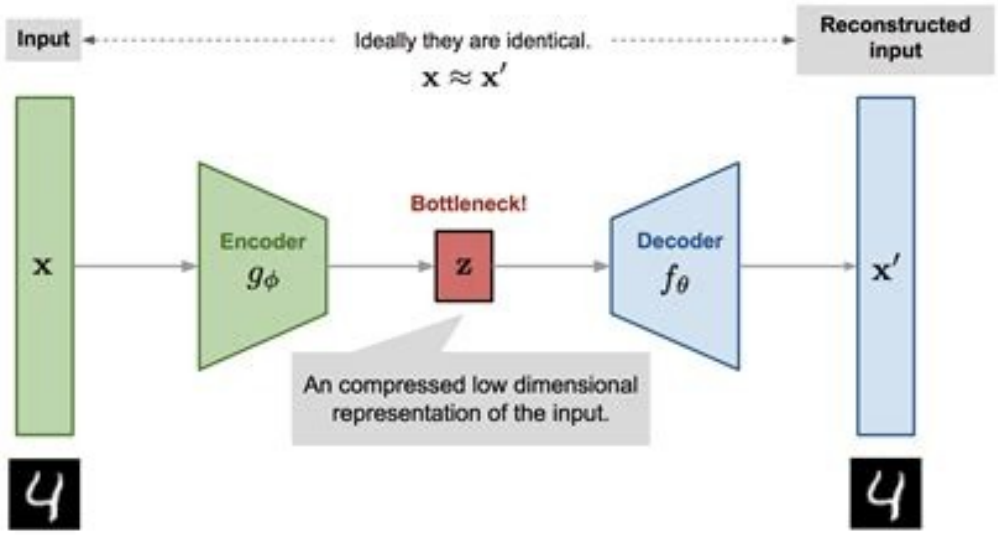
The Encoder

- Example
  - Input image of a dog
  - The encoder may encode
    - Ear shape
    - Fur color
    - Pose
    - Background type
  - But does not remember every pixel



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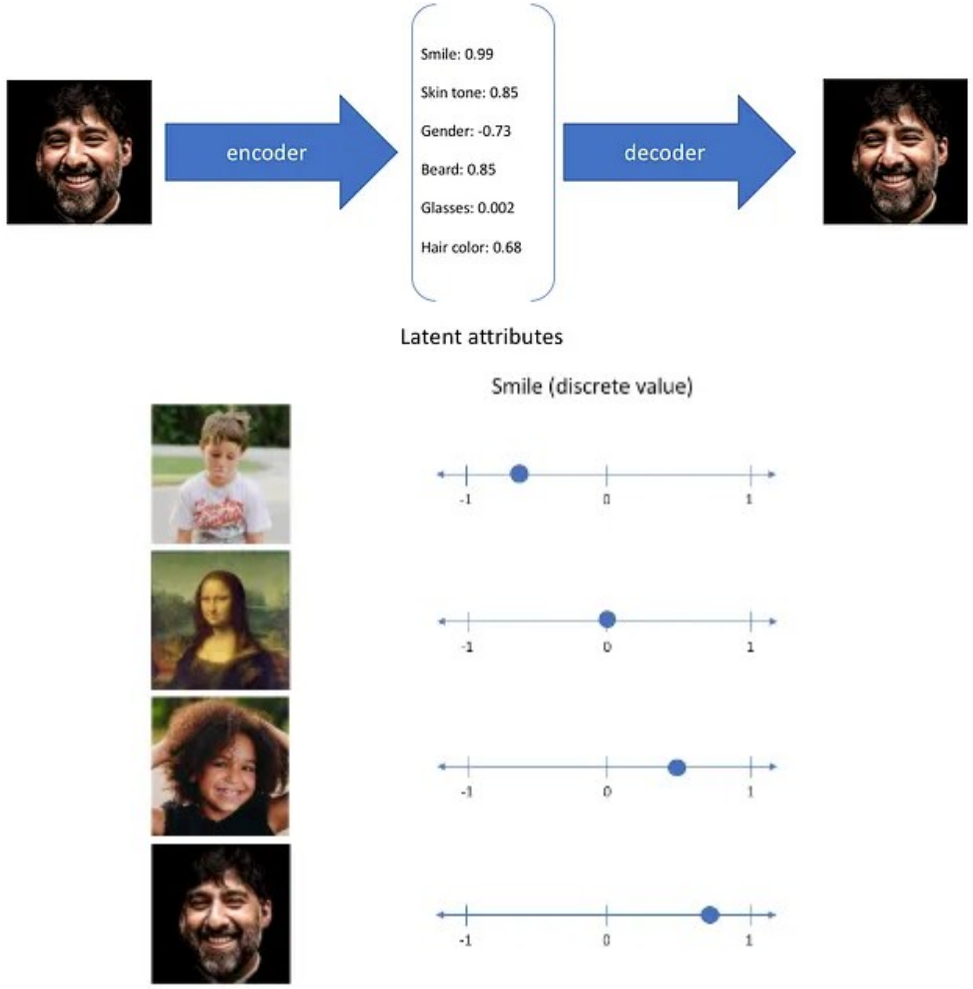
Latent Space

- Typically much smaller than the input
  - Is continuous (real-valued numbers)
  - Represents abstract features
- In the example shown, the face image is reduced to six features
  - For applications like removing noise, the autoencoder looks at the features defined in the latent space and ignores the noise
  - The autoencoder is not sensitive to small variations in the input



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Latent Space

- The latent space becomes structured
  - Because the network must:
    - Compress efficiently
    - Reconstruct accurately
  - It learns to organize latent space meaningfully
- This often produces
  - Smooth transitions
  - Clustering of similar inputs
  - Meaningful interpolation

```
z1 (smiling face)
z2 (neutral face)

Interpolate:

z = 0.5z1 + 0.5z2

→ Half-smiling face
```

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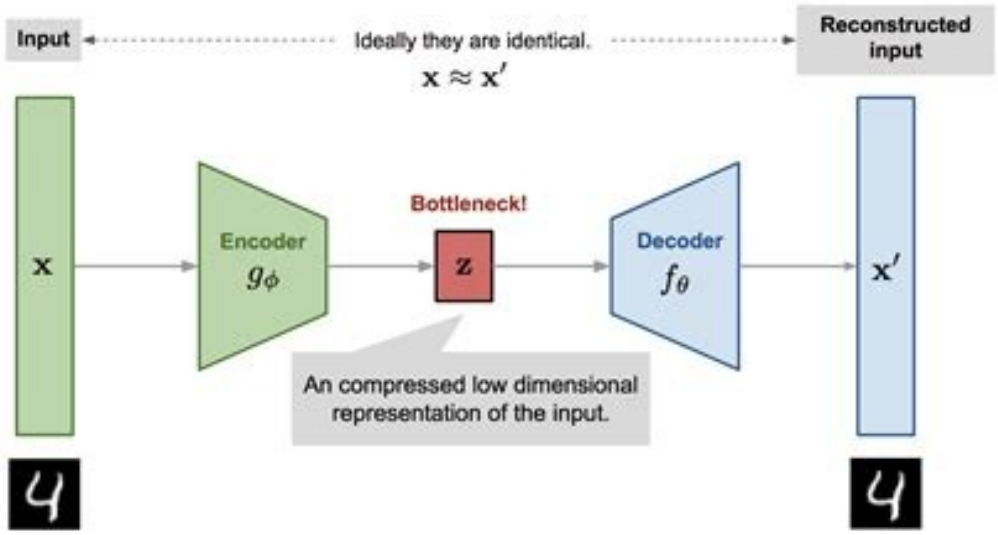
The Decoder

- The decoder reconstructs the input from the latent vector
- Mathematically where “f” is the decoder function
    - x’=f(z)
    - z = latent vector and x’ = reconstructed image
- The decoder asks:
  - “Give this compressed representation, how do I rebuild the original image?”
    - It expands abstract features back into full pixel detail



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Autoencoder Example

- Useing encoding and decoding to remove noise from an image

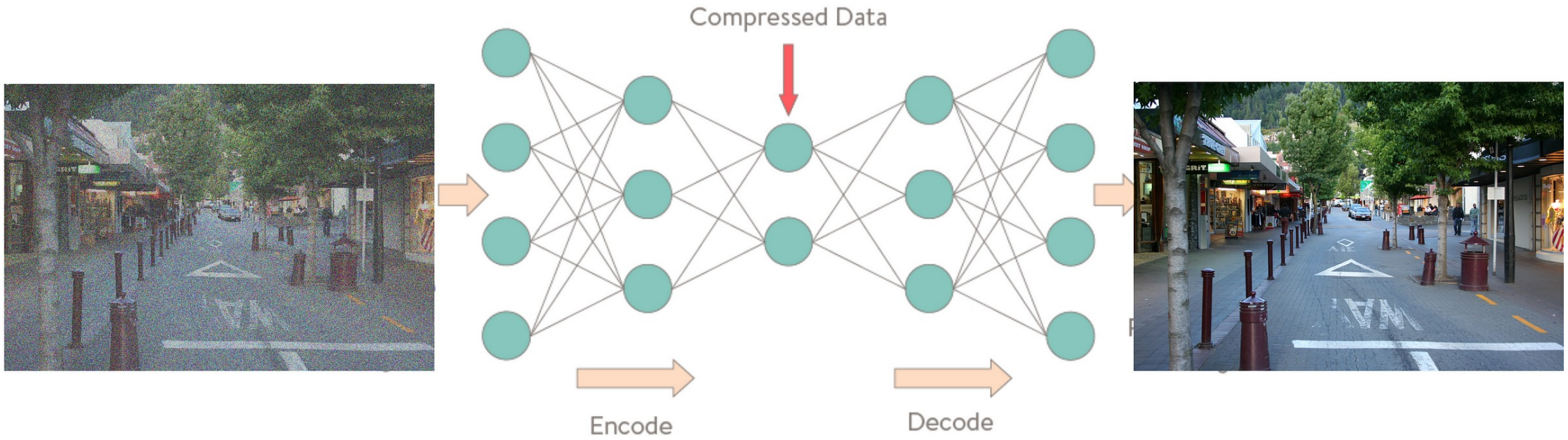

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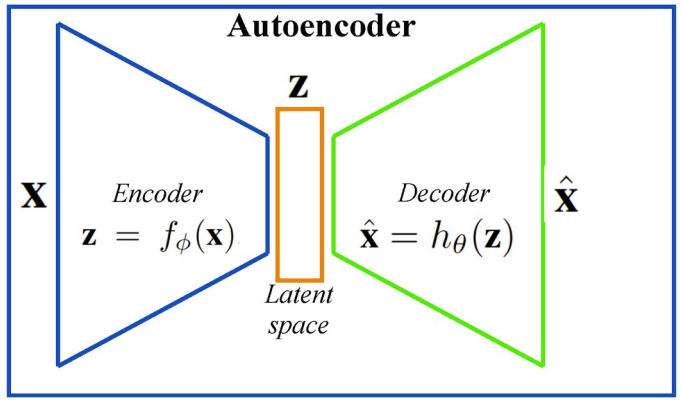
Image Credit: https://taus-epic.medium.com/dimensionality-reduction-with-autoencoders-27db23b71b12

Variational Autoencoder (VAE)

- A normal autoencoder
  - Learns to reconstruct inputs
  - Produces a latent vector
  - Minimizes reconstruction error
- But
  - The latent space is unstructured
  - Random sampling in latent space produces garbage
  - There is no guarantee latent vectors follow any distribution
  - So although it compresses data, it cannot reliably generate new samplesS
- We need something more for GAI



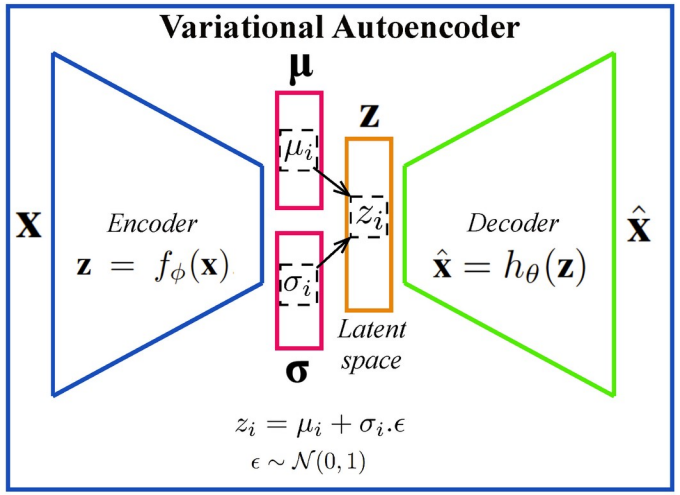


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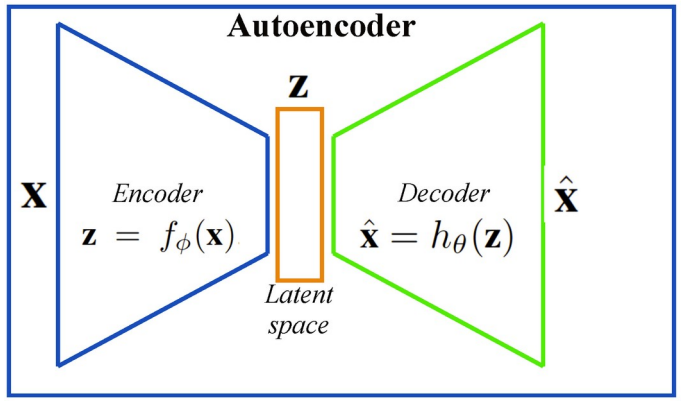
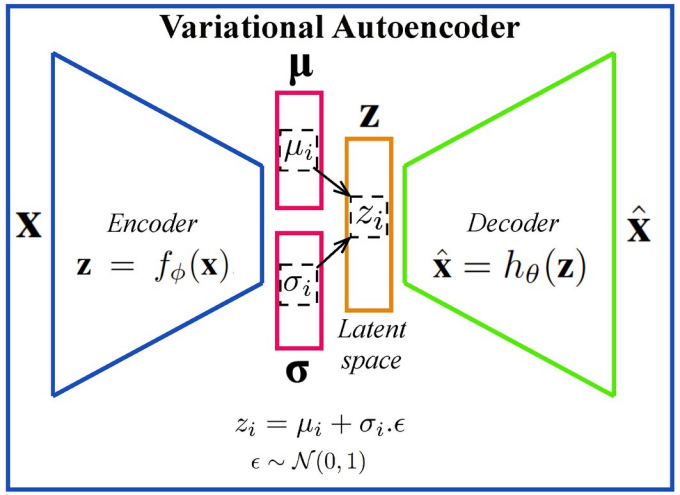
Image Credit:https://www.mdpi.com/2504-4990/2/3/20

Variational Autoencoder (VAE)

- A VAE turns the latent space into a structured probability space
  - Instead of mapping input to a single vector
    - z=f(x)
  - A VAE maps input to a distribution:
    - q(z|x)=N(μ(x),σ(x))
  - The encoder outputs
    - Mean (μ)
    - Variance (σ)
  - So each input becomes a region in latent space





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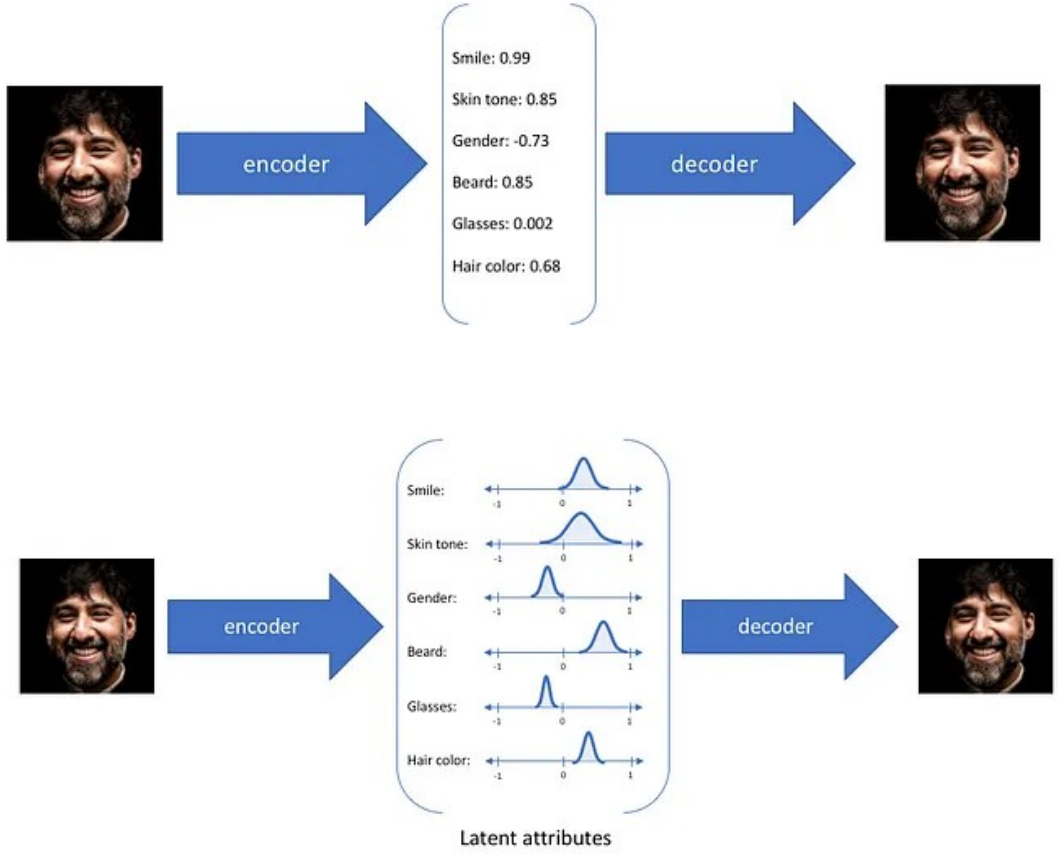
Latent Space Distribution

- With a VAE
  - The latent space becomes continuous
  - It becomes smooth
  - It becomes globally organized
- And most importantly
  - We can sample from a standard Gaussian
  - z∼N(0,I)
- Now the decoder will generate realistic outputs



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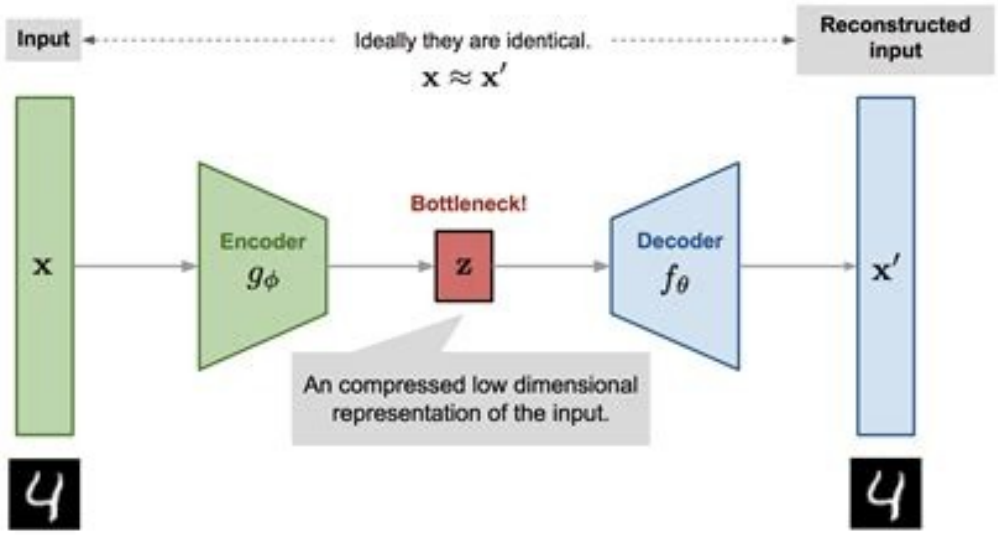
Importance of VAEs

- Introduced structured latent spaces
  - Modern generative models depend heavily on structured latent representations
- Made probabilistic generative modeling practical
  - VAEs connect deep learning with
    - Bayesian modeling
    - Variational inference
    - Approximate posterior learning
- Used inside modern diffusion systems



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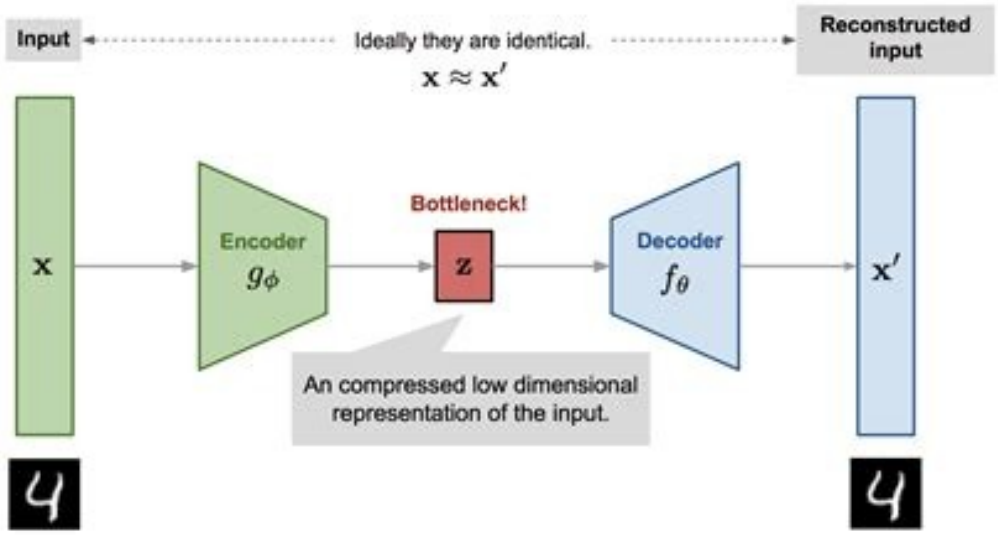
Importance of VAEs

- Without VAEs
  - Diffusion would be too expensive in pixel space
  - Consumer-level generation would not be feasible
  - VAEs power modern image generation
- VAEs are widely used for
  - Representation learning
  - Feature extraction
  - Generative modeling
  - Anomaly detection



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Representation Learning

- The process used by a model to automatically discovers useful features of raw data
  - Instead of manually defining features, the model learns them
- Raw data is messy
  - An image has
    - Pixel values
    - High dimensional
    - No explicit meaning
    - Not organized by “concept”
  - A learning system needs
    - Edges
    - Shapes
    - Objects
    - Semantics
    - Context

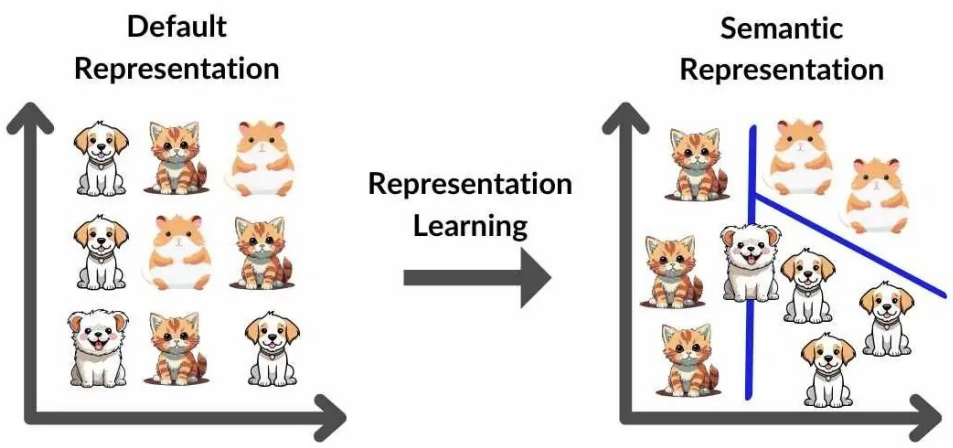

Image Credit: https://adinh26101.github.io/posts/representation-learning/

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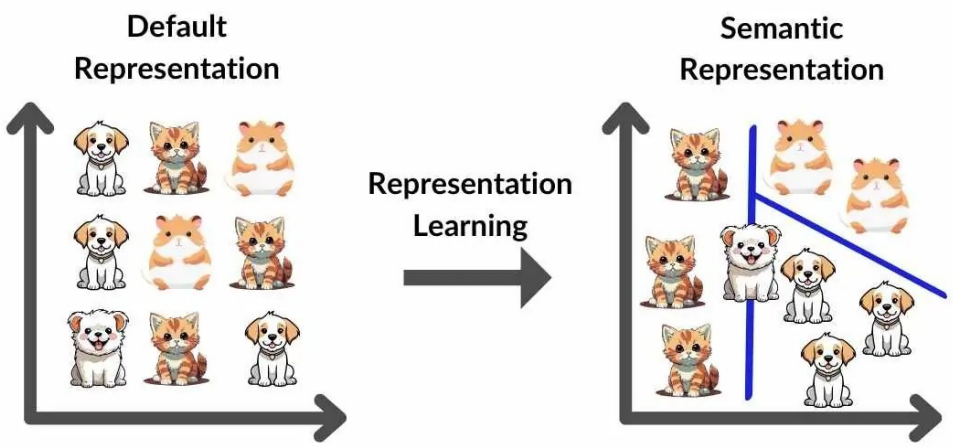
Representation Learning

- Traditional feature engineering
  - Human defines features
    - Edge detectors
    - Texture measures
    - Handcrafted descriptors
  - Then the model learns from these
- Representation learning, the model learns features automatically
  - Layers learn
    - Low-level patterns
    - Mid-level structure
    - High-level concepts
  - No manual feature engineering



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Diffusion Models

- VAE outputs are often blurry
- Because they optimize likelihood and reconstruction loss, which tends to average over possibilities
  - GANs solved sharpness,but introduced instability.
  - VAEs are stable but not sharp
- Can we get both stability and high realism?
  - The solution is diffusion models

Image Credit: https://www.researchgate.net/figure/Blurry-image-generated-by-VAE_fig2_375787474

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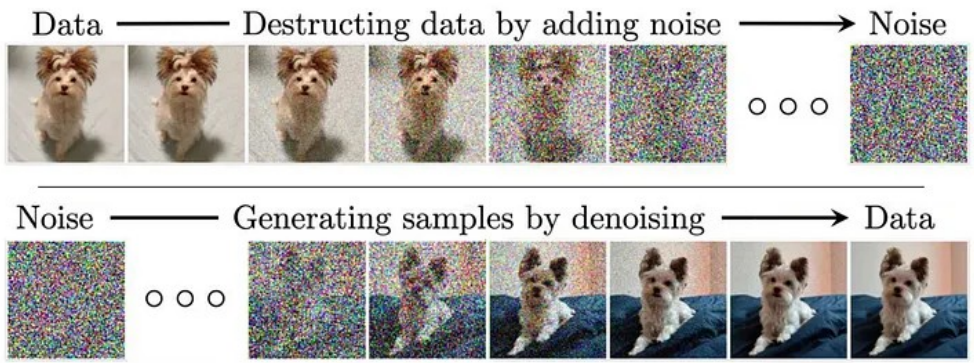
Diffusion Models

- Diffusion models take a different approach that GANs or VAE
- VAE approach
  - Encoding → sampling → decoding
- GAN approach
  - Generator vs discriminator
- Diffusion
  - Models lean how to reverse a gradual noising process
  - Essentially, erasing an image's features then restoring the erased features



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Training Diffusion Models

- For each real training image:
  - Take a clean image x_0
  - Randomly pick a timestep t
  - Add a known amount of Gaussian noise
  - Get a noisy version x_1
  - Repeat this for number of time steps
- Diffusion is learning to create a vector field over the latent space of images
  - Predicting noise is equivalent to learning the score function (in red)
  - The model learns the direction in which the data density increases. Where the real data clusters in high-dimensional latent space

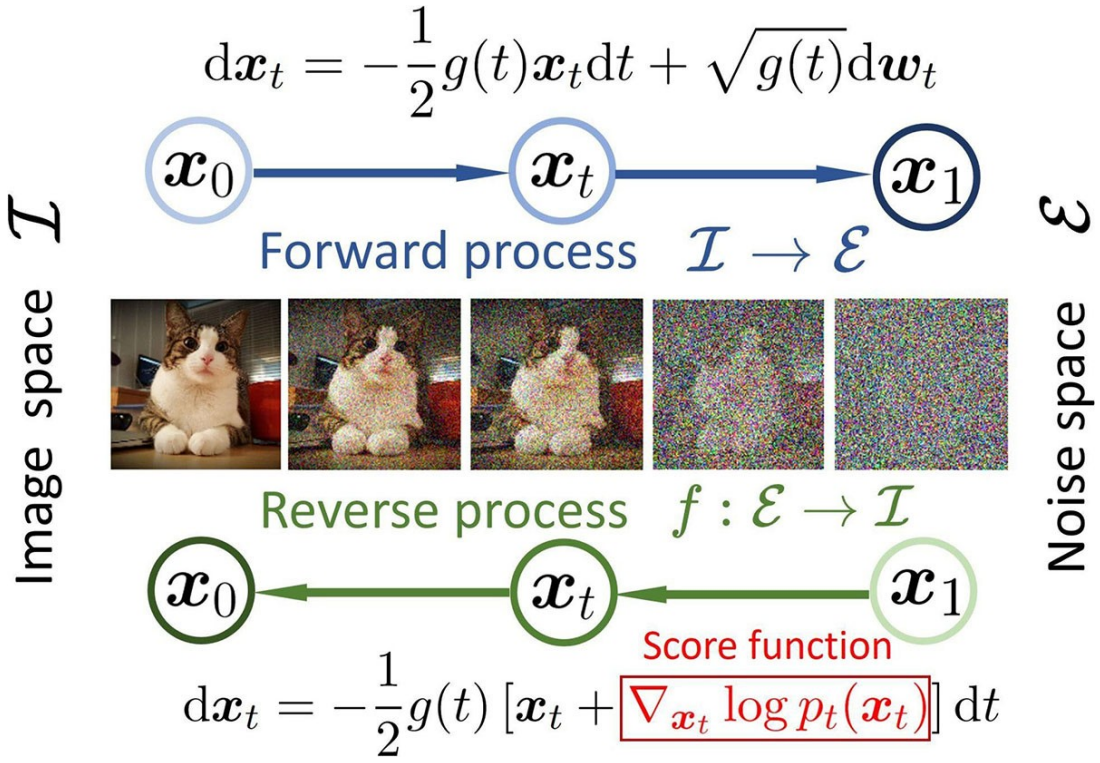

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Image Credit:https://www.siam.org/publications/siam-news/articles/generalization-of-diffusion-models-principles-theory-and-implications/

Training Diffusion Models

- By learning to remove noise at all noise levels, the model implicitly learns
  - What real images look like
  - What structure looks like
  - What edges look like
  - What textures look like
  - What objects look like
- To remove noise properly,
  - The model it must understand the underlying image structure

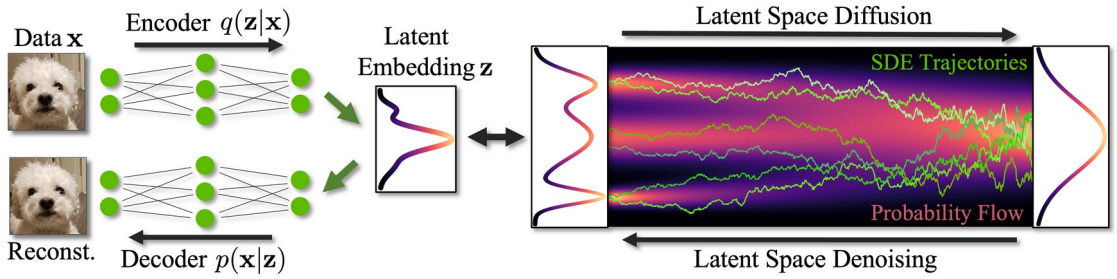

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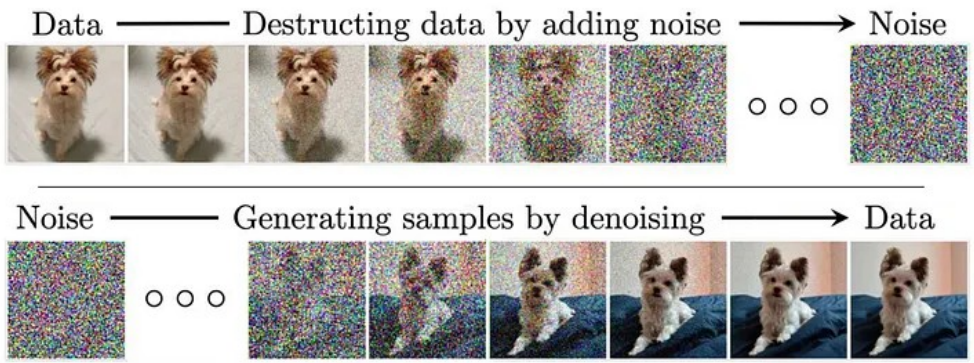
Image Generation

- Given a prompt, the denoising process used in training is followed
  - The starting point is random noise
  - The prompt provides a “direction” in the latent space
  - The generation phase removes noise that moves the image along the direction of the prompt



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Painter Analogy

- Your task is to teach students to paint using a diffusion process
- Training phase: learning restoration
  - You give the student a real masterpiece painting
  - You deliberately damage it slightly
    - Adding scratches, blur, stains, etc
  - Students are told “Restore this painting”
  - They try to fix it
    - You compare it to the original and evaluate how well it matches the original
  - Feedback improves their restoration skills

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Painter Analogy

- You repeat this process thousands of times
  - Sometimes the painting is lightly damaged
  - Sometimes it’s heavily damaged
  - Sometimes it’s almost completely ruined
- Students become very good at
  - Recognizing structure
  - Understanding composition
  - Understanding color relationships
  - Understanding texture and style

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Painter Analogy

- They are not memorizing paintings
  - They are learning the basics of what paintings are made up of
- Students have learned
  - How to move from disorder toward structure
  - What “real paintings” statistically look like

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Painter Analogy

- Generation phase
  - Students are now given a canvas covered in random paint splatter that looks like noise
  - They are given a prompt as to what the painting should be
  - Students start refining the splatter
    - They adjust shapes.
    - They remove randomness.
    - They add structure.
    - They impose symmetry and coherence
  - Each action makes the canvas
    - Slightly less random
    - Slightly more structured
  - After many small refinements:
    - A coherent painting emerges
    - They didn’t paint it in one jump
    - They gradually restored structure from chaos.

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Latent Diffusion

- The efficiency problem
- Diffusion requires
  - Many denoising steps
  - Large U-Net architecture (convolutional NN)
  - Heavy GPU compute processing
- Running diffusion in pixel space is slow and expensive

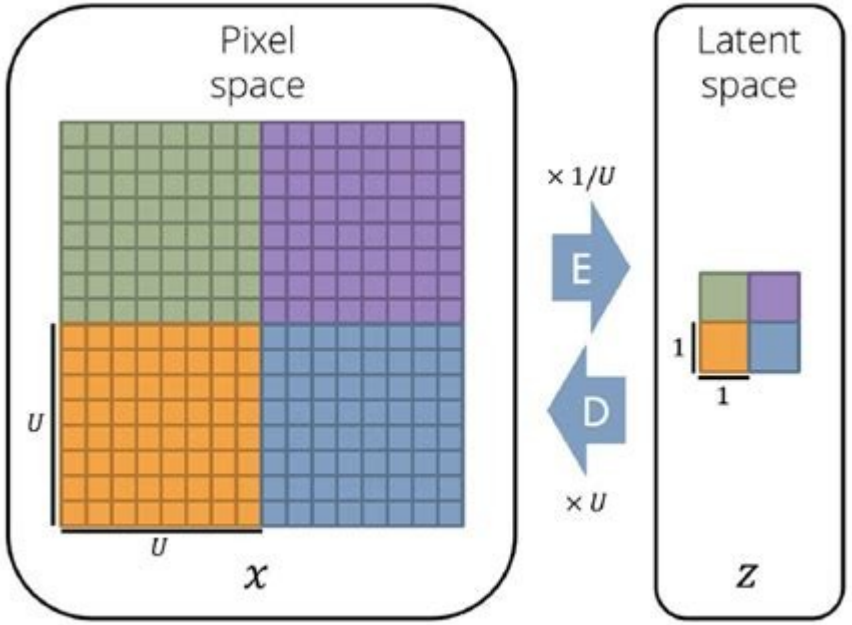

Image Credit: https://blog.metaphysic.ai/new-system-enables-tight-compositions-and-high-res-output-in-stable-diffusion/

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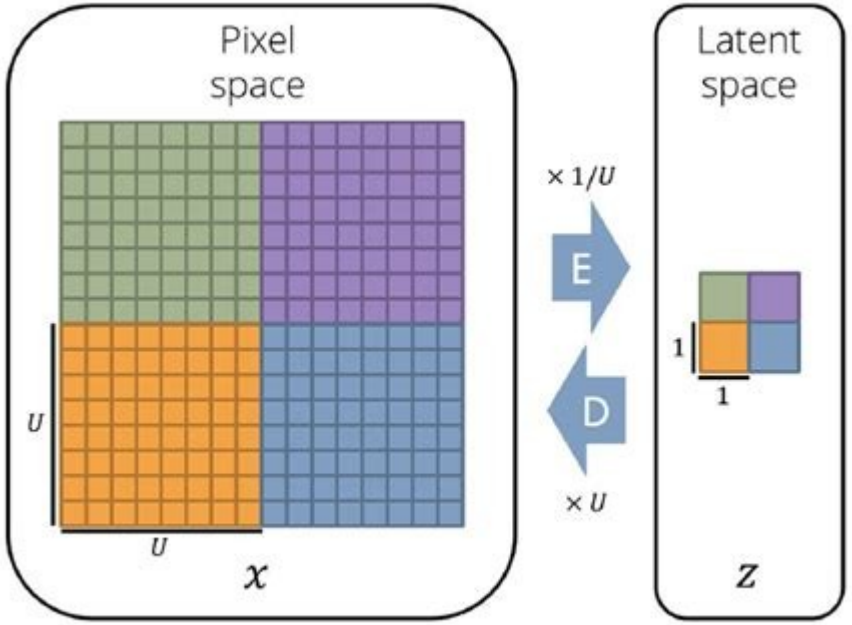
Latent Diffusion

- Stable Diffusion
  - Introduced applying diffusion in latent space rather than pixel space
- Latent diffusion
  - A VAE compresses the image
  - Diffusion operates in the compressed representation
  - The VAE decoder reconstructs pixels
- The encoder learns
  - Semantic structure
  - Object-level representation
  - Spatial coherence
- Diffusion works in a more meaningful space



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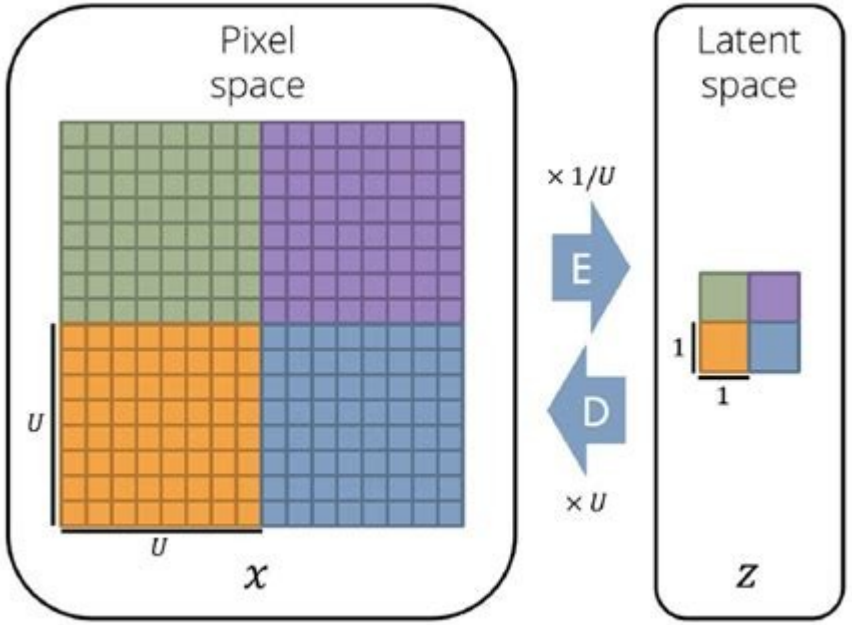
Latent Diffusion

- In pixel diffusion
  - Noise is added to pixels.
- In latent diffusion
  - Noise is added to latent vectors
- Computational efficiency
  - Faster training
  - Lower memory
  - Consumer GPU feasible
  - Practical deployment



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Multi-modal Training

- A modality is a type of data
  - Text, images, audio, video, etc
- A multi-modal model can process and relate multiple modalities
  - Understand text and images together
  - Describe an image in natural language
  - Generate an image from a text prompt
  - Answer questions about a picture

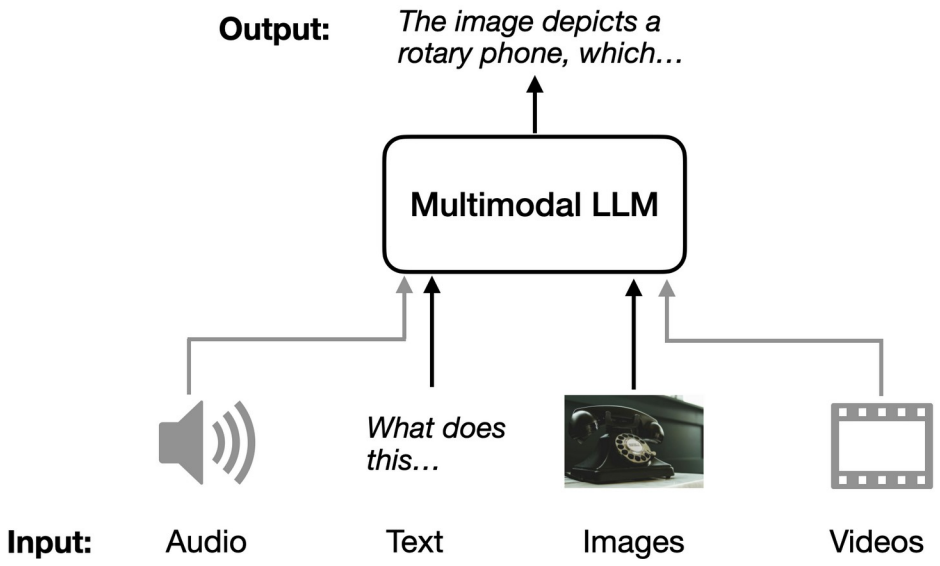

Image Credit https://sebastianraschka.com/blog/2024/understanding-multimodal-llms.html

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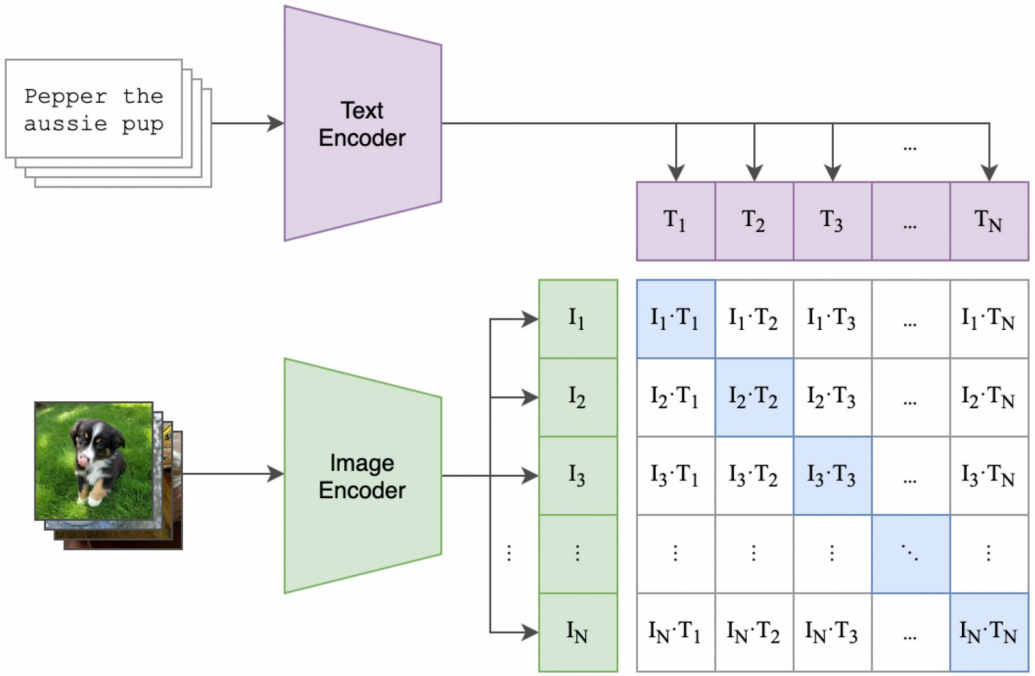
Multi-modal Training

- Training a multi-modal model
  - We want the model to understand that
    - The word “dog”
    - And a picture of a dog
    - Represent the same concept
- Shared embedding space
  - Instead of just Text → embedding space
  - The model trains a labeled image
    - Text → embedding space
    - Image → embedding space
  - For example
    - Text: "dog" → [0.12, -0.44, 1.03, ...]
    - Image of dog → [0.10, -0.40, 1.01, ...]



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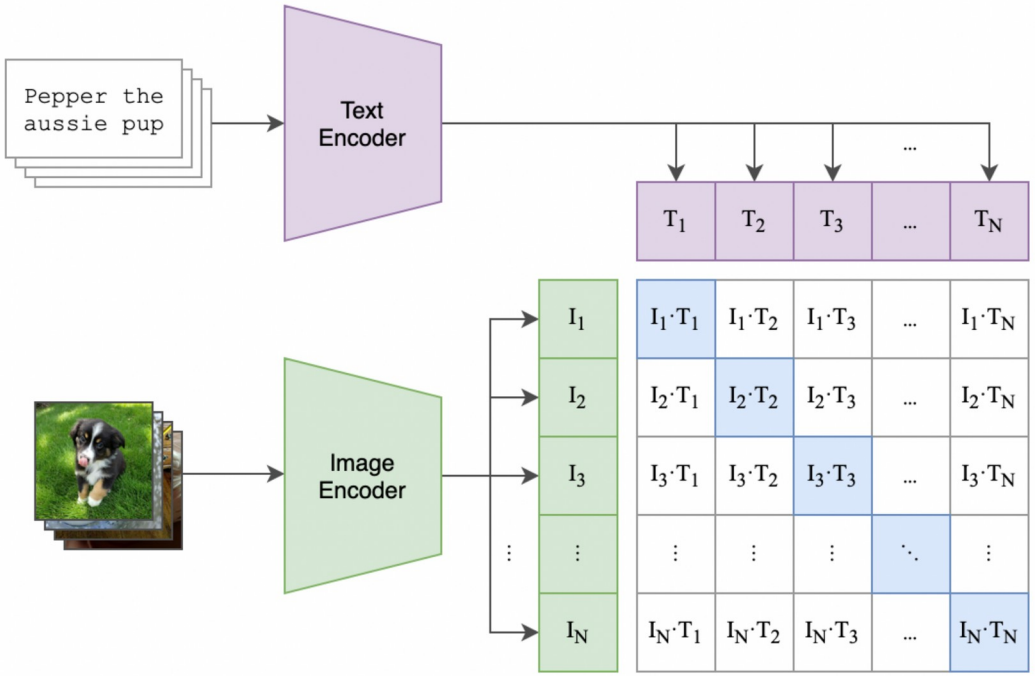
Multi-modal Training

- If two vectors are close, they represent the same concept
  - The paired image and text vectors are aligned, which means
    - To adjust the neural network weights so that matching image-text pairs produce similar vectors.
  - And
    - To adjust weights so that non-matching pairs produce dissimilar vectors
  - Alignment = optimizing distance relationships in vector space



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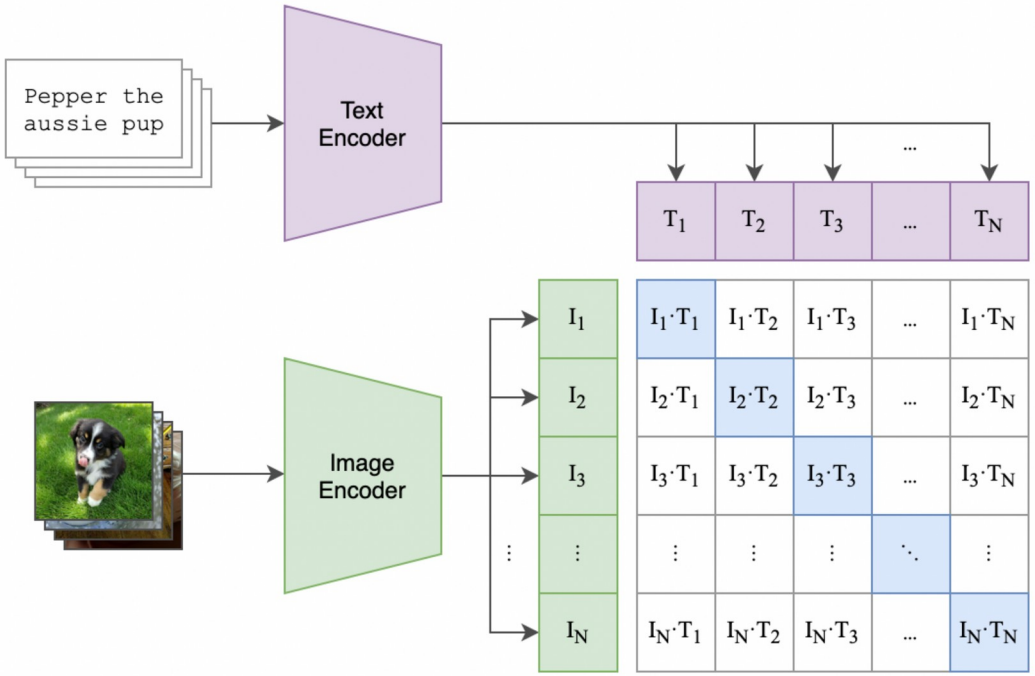
Multi-modal Training

- Step by step
  - Take a batch of image-text pairs
  - Encode them separately with 2 NNs
    - Image encoder → produces image vectors
    - Text encoder → produces text vectors
    - So we get

```
Image vectors:
I1, I2, I3, I4

Text vectors:
T1, T2, T3, T4
```



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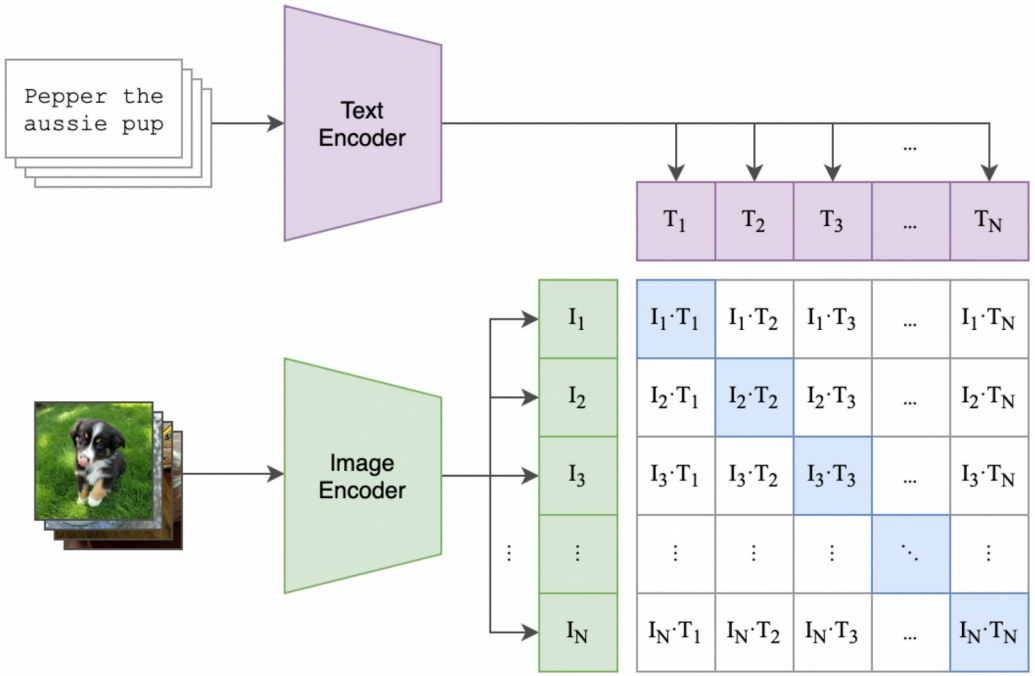
Multi-modal Training

- Step by step
  - Compute similarities
    - We compute similarity between every image and every text
  - We want
    - I1 close to T1
    - I2 close to T2
    - And far from others

	T1	T2	T3	T4
I1	0.91	0.12	0.08	0.05
I2	0.11	0.88	0.14	0.02
I3	0.07	0.09	0.85	0.10
I4	0.05	0.03	0.09	0.92



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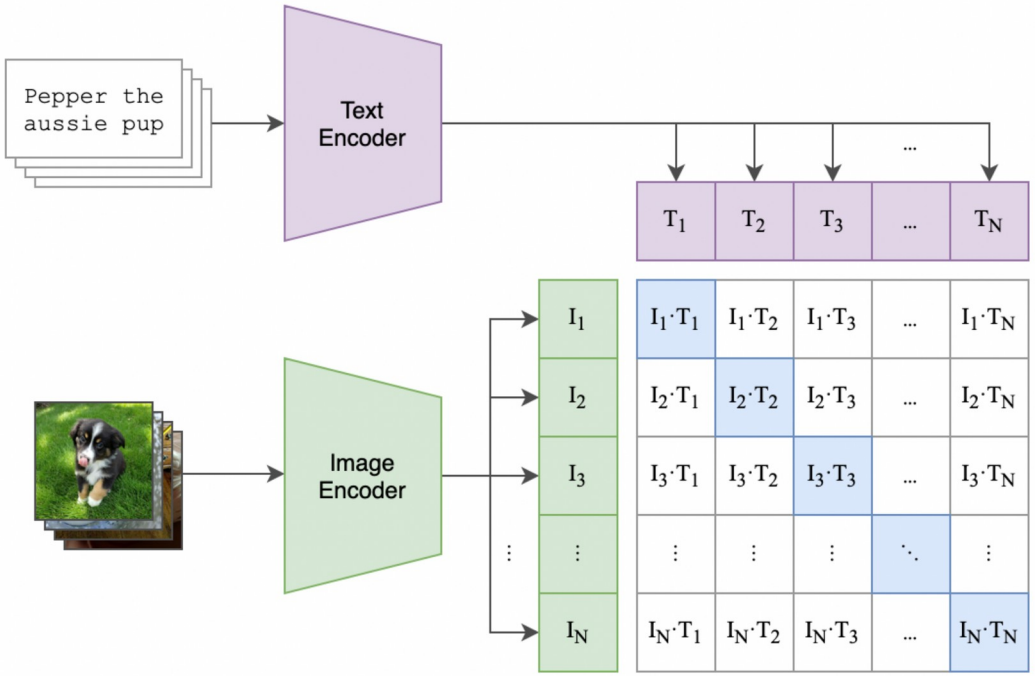
Multi-modal Training

- Step by step
  - Define the objective as a classification problems
    - For each image, the correct caption should have highest similarity
    - For each caption, the correct image should have highest similarity
  - For image I1
    - T1 should be classified as correct
    - T2, T3, T4 are negatives incorrect
  - Loss function penalizes
    - Wrong pairings being too close
    - Correct pairings being too far



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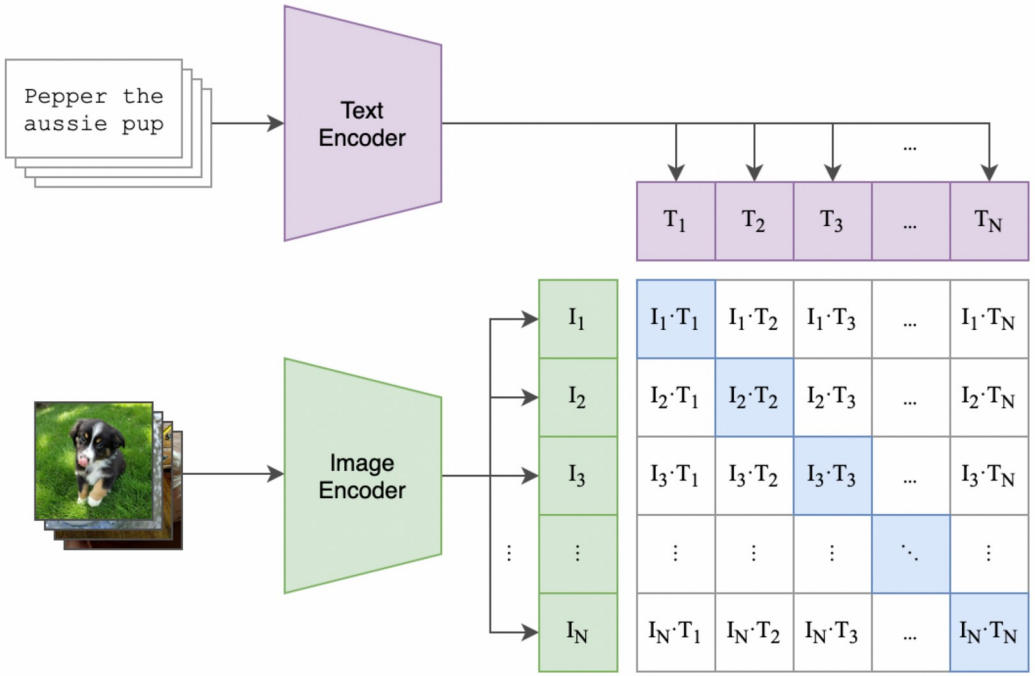
Multi-modal Training

- Step by step
  - Backpropagation adjusts
    - Image encoder weights
    - Text encoder weights
  - So that
    - Dog images move closer to dog-related words
    - Car images move closer to car-related words
    - Unrelated pairs move apart
  - Over millions of examples, structure emerges



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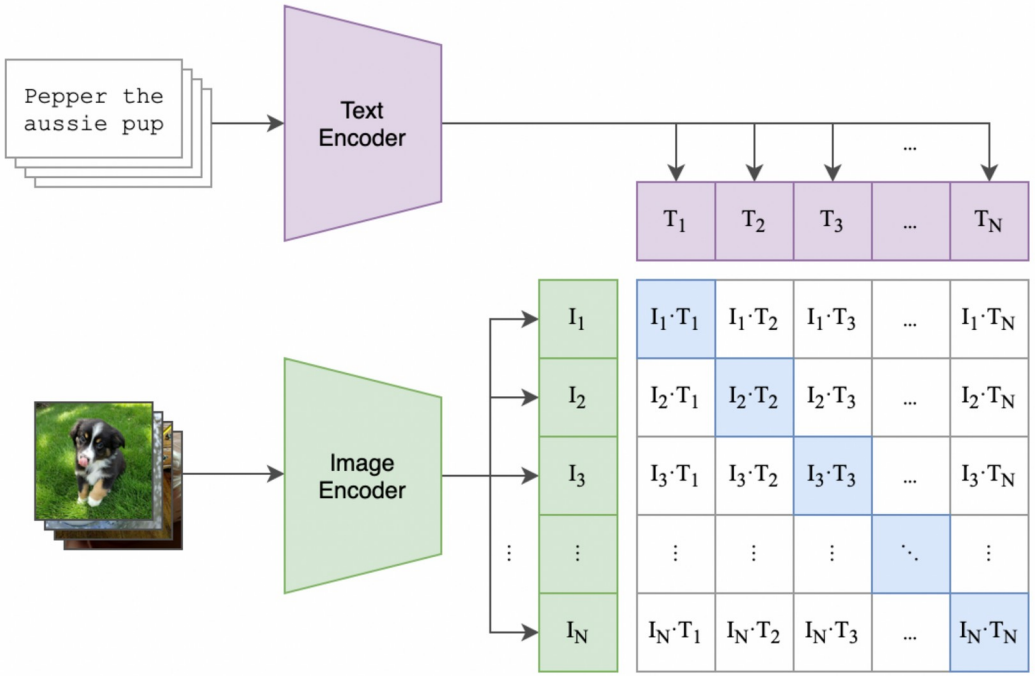
Multi-modal Training

- Why this works at scale
  - With millions/billions of image-caption pairs
  - The model learns statistical associations like
    - “Snow” → white textured background
    - “Sunset” → orange lighting
    - “Portrait” → centered human face
  - These patterns become encoded in the embedding space



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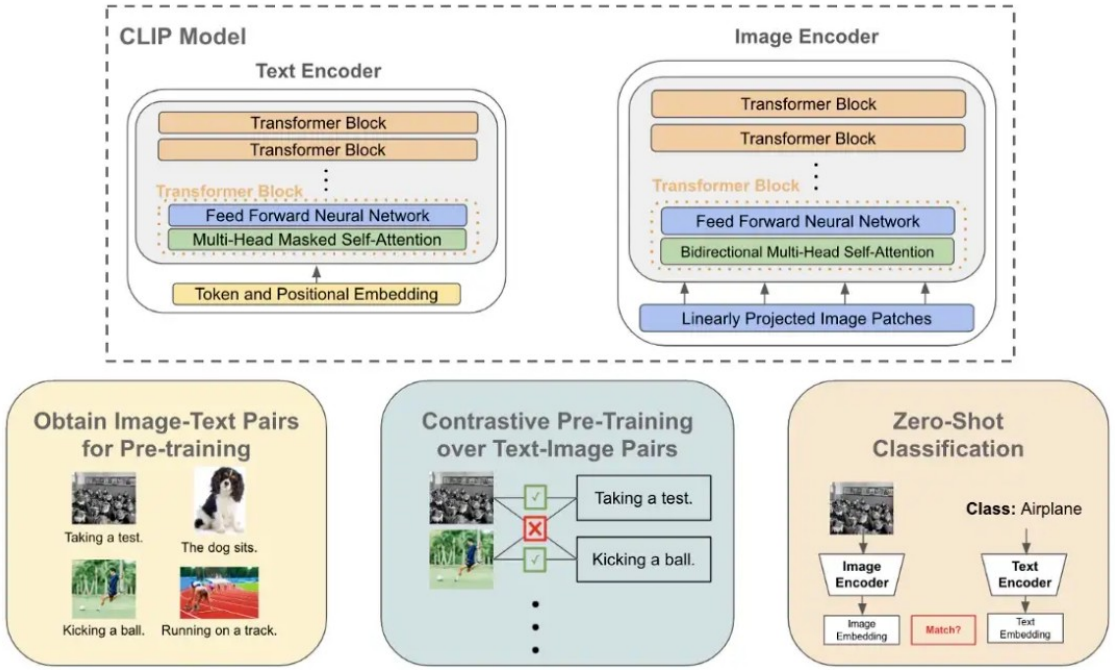
CLIP

- Contrastive language image pretraining
- CLIP is a model trained to
  - Understand the relationship between images and natural language descriptions
  - CLIP does not generate images
  - CLIP does not generate text
  - CLIP learns a shared semantic embedding space between images and text



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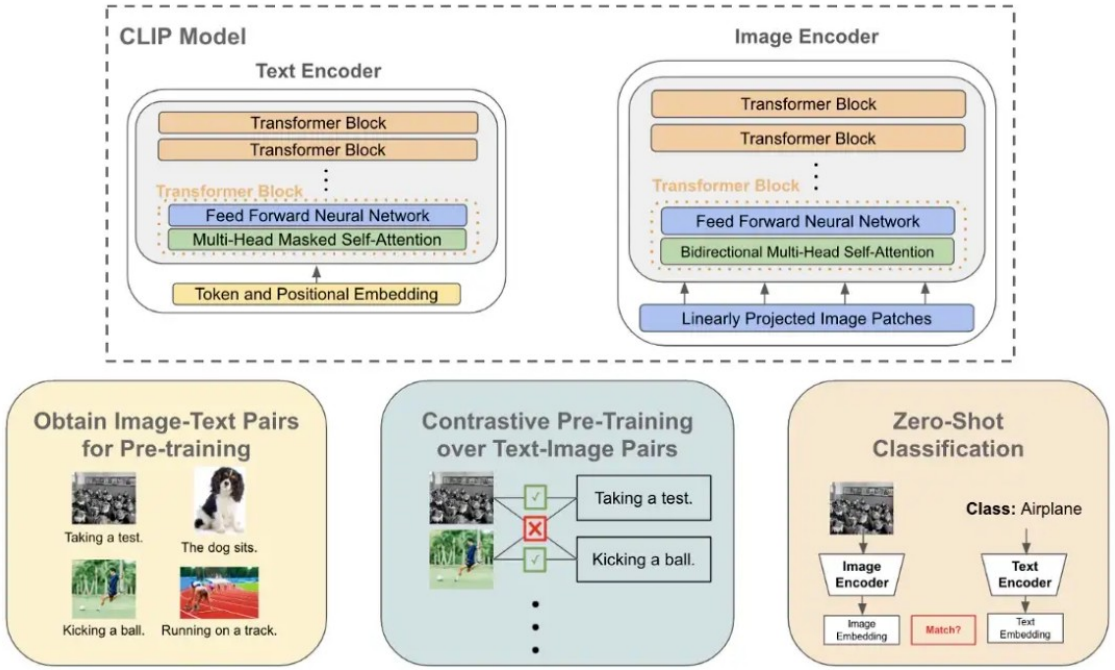
CLIP

- CLIP has two separate encoders
  - Image Encoder
    - Typically a CNN or Vision Transformer (ViT)
    - Input: image
    - Output: image embedding vector
  - Text Encoder
    - Transformer-based (similar to GPT/BERT-style encoders)
    - Input: text caption
      - *Output: text embedding vector*
  - They are separate networks
  - They are trained together



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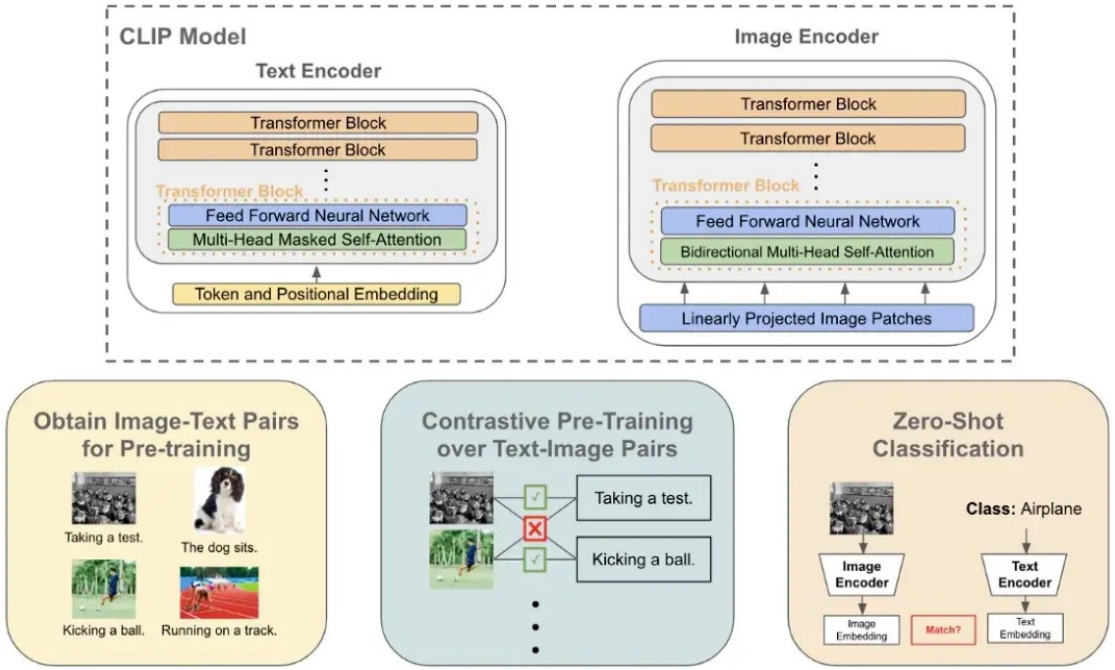
CLIP

- CLIP is trained on massive datasets of
  - Image plus caption pairs
  - Example
    - Image: Dog running on beach
    - Caption: “A dog playing on the beach”
  - Trained on millions or billions of such pairs
  - In each batch of N images and captions
    - Each image matches exactly one caption
    - CLIP computes similarity between all image text combinations
  - The model is trained contrastively so that
    - The correct image-caption pair has highest similarity
    - All other pairs have lower similarity



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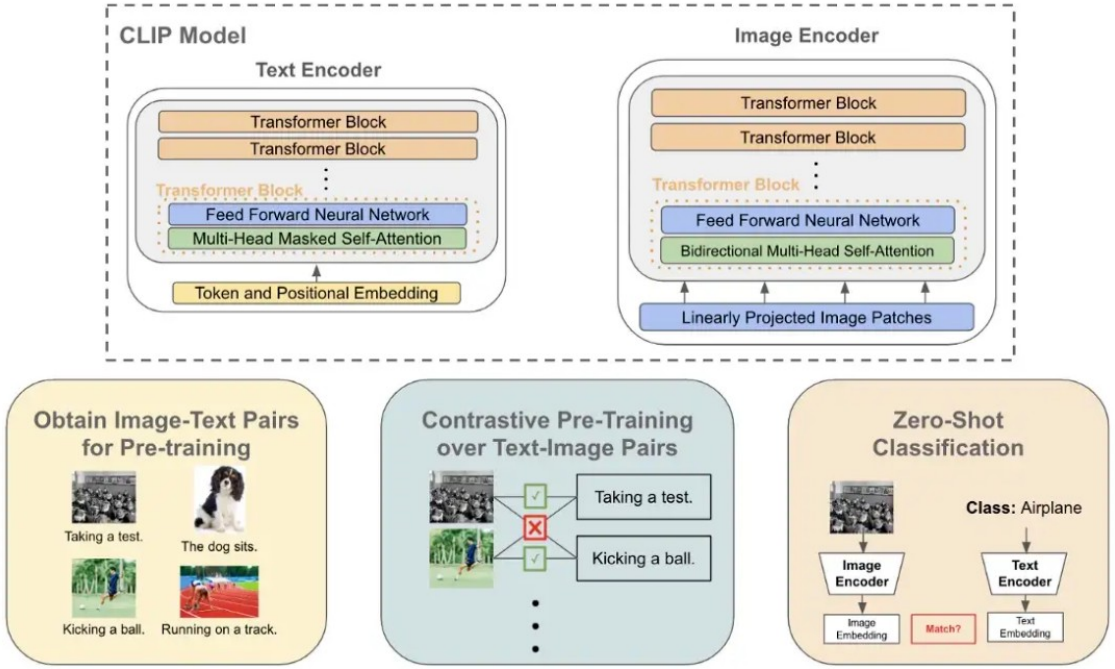
CLIP

- CLIP learns
  - High-level concepts like
    - Objects (dog, car, pizza)
    - Styles (oil painting, watercolor)
    - Attributes (red, shiny, blurry)
    - Relationships (on top of, next to)
  - Without manual labeling
  - It learns directly from natural language supervision
    - Specifically, the labels are the captions



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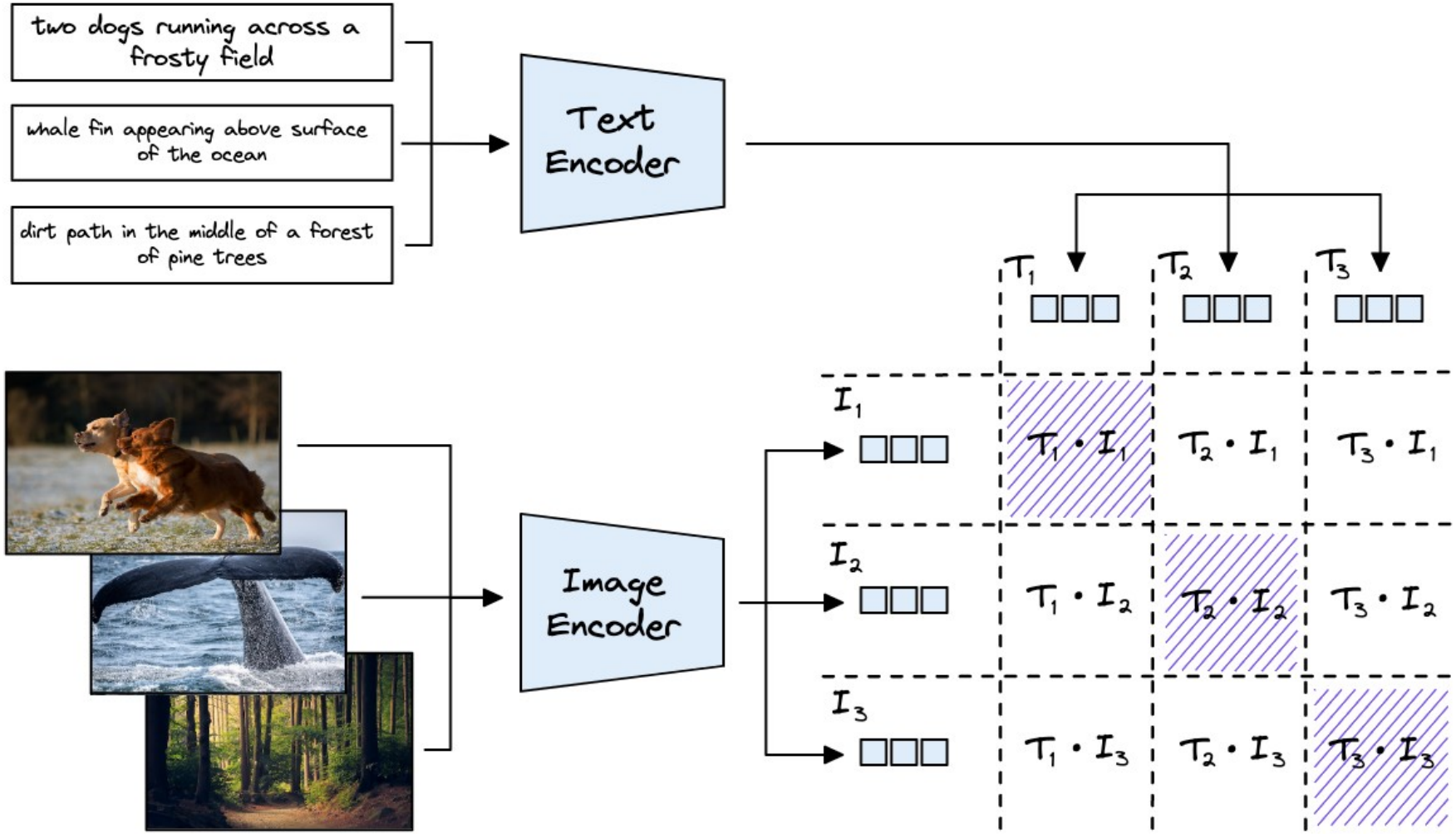
CLIP



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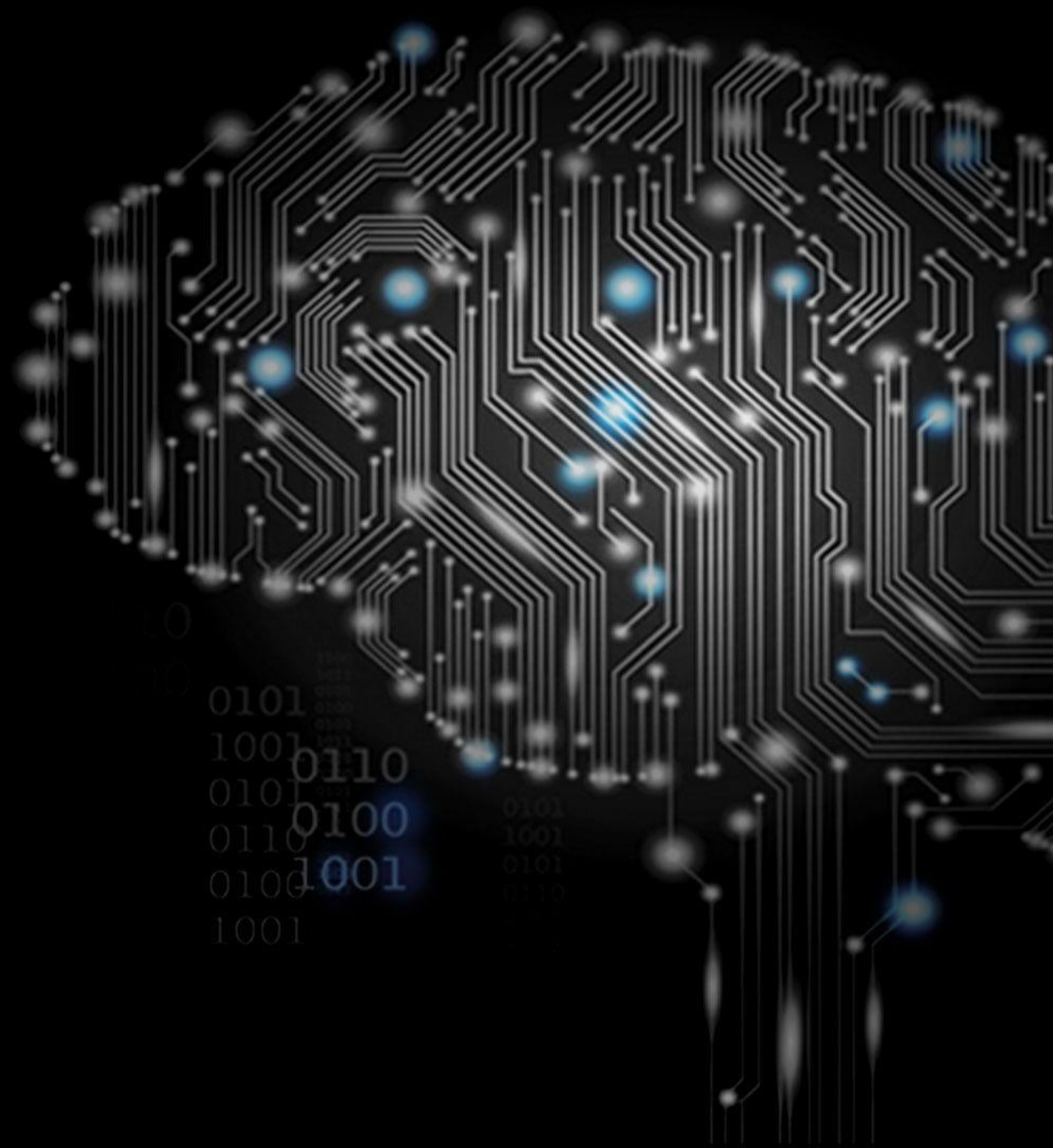

Questions?